C CodeCraft'26

# Emergency Rescue Priority System

Project Documentation – Scenario 1

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## 1. About the Project

A console-based C program that manages emergency rescue requests in a control room. Each request is categorised by severity and sorted so that Critical cases are handled before High and Normal cases. The program covers the full request lifecycle: registration, prioritisation, team assignment, status tracking, searching, summary reporting, and saving to a file.

## 2. Features

| Feature | Description |
|---|---|
| Register requests | Captures Request ID, Location, People Affected, Emergency Level and Required Resources, with input validation. |
| Priority categorisation | Maps Critical, High and Normal to priority 1, 2 and 3. Input is case-tolerant (e.g. critical or Critical). |
| Priority-based display | Sorts all requests with bubble sort so the most urgent appear first. |
| Rescue team assignment | Assigns a named rescue team to a request and sets its status to InProgress. |
| Status update | Updates a request to Pending, InProgress or Resolved, re-prompting on invalid input. |
| Search | Finds and displays a request's full details by Request ID. |
| Emergency summary | Reports total requests plus counts by emergency level and by status. |
| File storage | Saves all request details to rescue.txt in a pipe-delimited format. |

## 3. Data Structure

Each request is stored in a RescueRequest structure; up to 100 requests are held in an array of structures.

| Field | Type | Purpose |
|---|---|---|
| requestID | char[20] | Unique request identifier (e.g. ER102) |
| location | char[50] | Location of the emergency |

| Field | Type | Purpose |
|---|---|---|
| people | int | Number of people affected |
| emergency | char[20] | Critical, High or Normal |
| resources | char[50] | Resources required (e.g. Ambulance) |
| assignedTeam | char[30] | Assigned rescue team (default: NotAssigned) |
| status | char[20] | Pending, InProgress or Resolved |
| priority | int | 1 = Critical, 2 = High, 3 = Normal |

## 4. Program Workflow

- Enter the number of requests (1–100) and the details of each.
- Priority is assigned automatically from the emergency level.
- Requests are sorted and displayed in priority order.
- A team is assigned to a chosen request by Request ID.
- The status of a chosen request is updated.
- A request is searched by ID and its details are shown.
- The emergency summary is generated.
- All records are written to rescue.txt.

## 5. Technologies and Concepts Used

- C programming (GCC compiler)
- Structures and arrays of structures
- String handling (strcmp, strcpy)
- Bubble sort for priority ordering
- Linear search by Request ID
- File handling (fopen, fprintf, fclose)

## 6. How to Run

**Compile:**

```
gcc rescue.c -o rescue
```

**Run on Windows:**

```
.\rescue.exe
```

## 7. Sample Input and Output

Sample request (from the problem statement):

| Field | Value |
|---|---|
| Request ID | ER102 |

| Field | Value |
|---|---|
| Location | Sathyamangalam |
| People | 15 |
| Emergency | Critical |
| Resources | Ambulance |
| Status | Pending |

Output file format (rescue.txt):

```
ID|Location|People|Emergency|Resources|Team|Priority|Status
ER102|Sathyamangalam|15|Critical|Ambulance|NotAssigned|1|Pending
```

## 8. Current Limitations

- Saved requests are not reloaded when the program restarts.
- Multiple rescue teams and their availability are not managed; the team name is entered manually.
- The program runs as a single sequential flow rather than a repeating menu.
- Capacity is fixed at 100 requests (static array).
- Duplicate Request IDs are not rejected, and team names cannot contain spaces.

## 9. Future Improvements

- Reload records from rescue.txt on startup.
- Team array with availability flags, freeing a team when a request is resolved.
- Menu-driven interface with a loop.
- Dynamic memory allocation (malloc/realloc).
- Additional sorting criteria (e.g. people affected) and response-time statistics.

## 9. GIT HUB REPOSITARY LINK

https://github.com/KIRANKUMARV2007/Emergency-Rescue-Priority-System-for-CODE_CRAFT

## 10. OUTPUT OF PROGRAM

```
PS C:\Users\ Debug Console (Ctrl+Shift+Y) AFT> gcc rescue.c -o test
PS C:\Users\vkira\Downloads\C_CODECRAFT> .\test.exe
===== EMERGENCY RESCUE PRIORITY SYSTEM =====
Enter number of rescue requests (1-100): 3

Enter details for Request 1
Request ID: ET101
Location: Sathy
Number of people affected: 14
Emergency type (Critical/High/Normal): Normal
Required resources: WheelChair

Enter details for Request 2
Request ID: ET102
Location: Gobi
Number of people affected: 22
Emergency type (Critical/High/Normal): High
Required resources: Ventilator and ECG

Enter details for Request 3
Request ID: ET103
Location: Gavundampadi
Number of people affected: 21
Emergency type (Critical/High/Normal): Critical
Required resources: Ambulance allocation and Xray,ECG machines

===== REQUESTS SORTED BY PRIORITY =====

ID: ET103
Location: Gavundampadi
People affected: 21
Emergency: Critical
Resources: Ambulance allocation and Xray,ECG machines
Priority: 1
Assigned Team: NotAssigned
Status: Pending

ID: ET102
Location: Gobi
People affected: 22
Emergency: High
Resources: Ventilator and ECG
Priority: 2
Assigned Team: NotAssigned
Status: Pending

ID: ET101
Location: Sathy
People affected: 14
Emergency: Normal
Resources: WheelChair
Priority: 3
Assigned Team: NotAssigned
Status: Pending

Enter Request ID to assign a team: ET103
Enter rescue team name (no spaces): TEAM-A01
Team assigned successfully!

Enter Request ID to update status: ET102
Enter status (Pending/InProgress/Resolved): InProgress
Status updated successfully!

Enter Request ID to search: ET101

===== REQUEST FOUND =====
ID: ET101
Location: Sathy
People affected: 14
Emergency: Normal
Resources: WheelChair
Team: NotAssigned
Status: Pending
Priority: 3
```

```
===== EMERGENCY SUMMARY =====
Total requests: 3
Critical: 1
High: 1
Normal: 1
Pending: 1
In Progress: 2
Resolved: 0

All requests saved to rescue.txt successfully!
PS C:\Users\vkira\Downloads\C_CODECRAFT>
```

## 11. CONCLUSION

The Emergency Rescue Priority System demonstrates how C programming can be used to organize emergency information and support priority-based rescue request management. By combining structures, arrays, string handling, sorting, searching, and file handling, the project provides a functional foundation for tracking emergency requests.

The implementation covers all seven core requirements of Scenario 1: request registration, priority assignment, priority-based display, team assignment, status updates, searching, and emergency summary generation. It also includes file storage from the bonus feature list. With further development in persistent data loading, team availability management, and flexible user interaction, the system could be extended into a more comprehensive emergency coordination application.